QF1100
AY25/26 Sem 2
by ngmh

Theory of Interest

Principal Amount Sum of money for investment.
Interest Time value of money.
Accumulation Function Accumulated value of $\$1$ at time t, with $a(0) = 1$.
Simple Interest Linear Growth: $a(t) = 1 + rt$.
Compound Interest Exponential Growth: $a(t) = (1 + r)^t$.
Frequency of Compounding If an interest is paid p times a year, p is the frequency of compounding, $r^{(p)}$ is the nominal interest rate, and the interest paid each time is $\frac{r^{(p)}}{p}$.
Effective Annual Interest Rate $a(1) = 1 + r_e = (1 + \frac{r^{(p)}}{p})^p$, with accumulation function $a(t) = (1 + r_e)^t = (1 + \frac{r^{(p)}}{p})^{pt}$. The effective interest rate is always greater than the nominal rate.
Equivalent Interest Rates Two nominal interest rates are equivalent iff they yield the same accumulation amount over a year, i.e. $1 + r_e$ is equivalent.
Continuous Compounding For a positive interest rate, $lim_{x\to\infty}(1 + \frac{r}{x})^x = e^r$. Interest is compounded continuously when the frequency of compounding tends to infinity. The accumulation function is $a(t) = e^{r^{(\infty)}t}, t \geq 0$.
Force of Interest The force of interest, is $\delta(t) = \frac{a'(t)}{a(t)} = \frac{d}{dt}[\ln(a(t))]$.

- When $\delta(t) = \delta, a(t) = e^{\delta t}$, the continuously compounded rate of interest.
- For simple interest, $\delta(t) = \frac{r}{1+rt}$.
- For compound interest, $\delta(t) = \ln(1 + r)$.

From definition, $a(s,t) = \exp(\int_s^t \delta(u)du)$. This implies the principle of consistency: $a(t_0, t_n) = a(t_0, t_1)a(t_1, t_2)...a(t_{n-1}, t_n)$.

Present Value

Present Value The present value of a payment c at time $t \geq 0$ is $\frac{c}{a(t)}$. Finding the present value of a future payment is known as discounting.
Present Value of Cash Flow Stream For a cash flow which is a series of payments, the present value $PV(C) = \sum_{i=1}^n \frac{c_i}{a(t_i)}$. If the effective annual interest rate is r, then $PV(C) = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i}}$.
Time Value The time value of a cash flow $TV_t(C) = PV(C) \cdot a(t)$. For $0 < s < t, TV_t(C) = \frac{a(t)}{a(s)} \cdot TV_s(C)$. If the effective annual interest rate is r, then $TV_t(C) = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i - t}}$ and $TV_t(C) = TV_s(C) \cdot (1 + r)^{t-s}$.
Principle of Equivalence Two cash flow streams are equivalent iff they have the same present value.
Net Present Value Considers PV of all negative and positive values, measure of value created today by undertaking investment.
Internal Rate of Return Non-negative root r of equation of value where $PV = 0$, a.k.a yield or IRR.
Unique IRR If unique, the IRR is the rate of return on investment. This occurs when $c_0 < 0, c_k \geq 0, c_0 + c_1 + ... + c_n > 0$. Otherwise, if the equation of value has degree n, then there are at most n distinct solutions.
Newton-Raphson There is no closed form solution for $n \geq 5$. First use IVT to pin down to a small interval where sign changes followed by choosing an initial guess. Then $\alpha_1 = \alpha_0 - \frac{f(\alpha_0)}{f'(\alpha_0)}$.
Using IRR IRR can be used to compare investments, just like NPV, higher is better.

Annuities

Annuity Series of payments made at regular intervals, cash flow stream with evenly spread payments.
Ordinary Annuity Payments are made at the end of each period, a.k.a annuity immediate or annuity in arrears. $PV = \sum_{t=1}^n \frac{A}{(1+r)^t} = \frac{A}{r}[1 - \frac{1}{(1+r)^n}]$.
Annuity Due Payments are made at the beginning of each period. $PV = \sum_{t=0}^{n-1} \frac{A}{(1+r)^t} = \frac{A(1+r)}{r}[1 - \frac{1}{(1+r)^n}]$. $PV(OA) = \frac{1}{1+r}PV(AD)$.
Time Value of Annuities $TV_t = PV \cdot (1 + r)^t$.
Perpetual Annuity Pays a fixed sum forever. If Due, $PV = \sum_{t=0}^\infty \frac{A}{(1+r)^t} = \frac{A(1+r)}{r}$. If Ordinary, $PV = \sum_{t=1}^\infty \frac{A}{(1+r)^t} = \frac{A}{r}$.

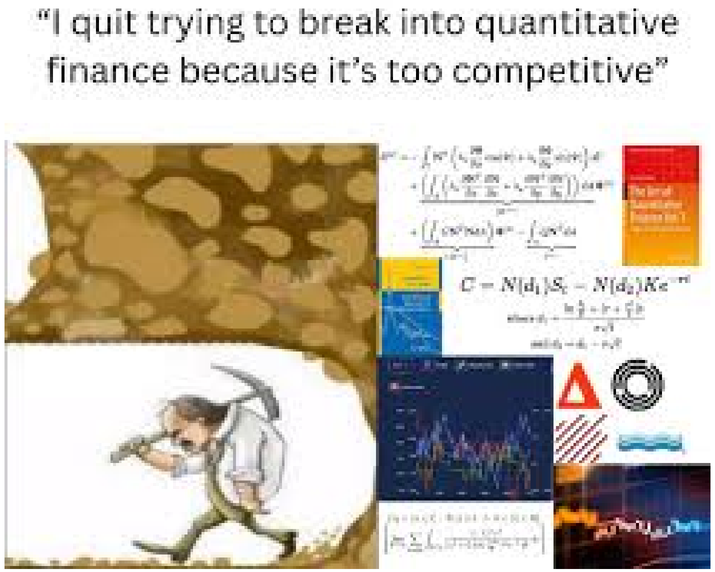
Loans

Loan Repaid by series of installment payments at periodic intervals. Cash flow is $+L, -A, -A, ...$
Refinancing Revise terms of existing loan, based on balance amount. The balance amount is the time value of the remaining unpaid payments.
Prospective Method Consider cash flows for $t > m$, then $L_{Bal} = \frac{A}{1+r} + ... + \frac{A}{(1+r)^{n-m}} = \frac{A}{r}[1 - \frac{1}{(1+r)^{n-m}}]$.
Retrospective Method Consider cash flows for $t \leq m$, then $L_{Bal} = L(1 + r)^m - \frac{A}{r}[1 - \frac{1}{(1+r)^m}](1 + r)^m$.
Number of Payments Find number of payments, loan will be repaid with $\lfloor n \rfloor$ full payments of A then a last payment of $B < A$.

Bonds

Bond Written contract between issuer and investors
Face / Par Value F, Amount based on which periodic interest payments are computed.
Redemption / Maturity Value R, Amount to be repaid at the end of the loan.
Maturity / Redemption Data Date on which loan will be fully repaid.
Coupon Rate c, Bond's interest payments, as a percentage of par value, made to investors at regular intervals during term of loan.
Other Symbols P: current price of a bond, m: number of coupon payments per year, n: total number of coupon payments, λ: nominal yield.
Law of One Price Price of bond equals to PV of cash flow consisting of all coupon payments and redemption value at maturity, calculated at yield λ.
Coupon Bonds Cash flow is made up of initial $-P$ and coupon payments of cF/m at multiples of $t = 1/m$, with redemption or maturity value $cF/m + R$ at time of last payment.
Valuing Coupon Bond
$P = \frac{R}{(1+\lambda/m)^n} + \sum_{i=1}^n \frac{cF/m}{(1+\lambda/m)^i} = \frac{R}{(1+\lambda/m)^n} + \frac{F\cdot c/m}{\lambda/m}(1 - \frac{1}{(1+\lambda/m)^n})$, made up of PV of redemption value and PV of coupon stream. Typically, $F = R$, then $P = F[\frac{1}{(1+\lambda/m)^n} + \frac{c/m}{\lambda/m}(1 - \frac{1}{(1+\lambda/m)^n})]$. Alternatively, $P = F + F(\frac{c-\lambda}{\lambda})[1 - \frac{1}{(1+\lambda/m)^n}]$.
Bond Purchase Premium: $P > F, c > \lambda$, Par: $P = F, c = \lambda$, Discount: $P < F, c < \lambda$.
Makeham Formula PV of face value at maturity is $K = \frac{F}{(1+\lambda/m)^n}$. $P = K + \frac{c}{\lambda}(F - K)$. Also, $P_{k+1} = P_k(1 + \lambda/m) - cF/m$, made up of accrued interest and coupon payment, after the kth payment.
Zero Coupon Bonds Do not pay coupons, cash flow is maturity value at the end. Then $P = \frac{R}{(1+\lambda)^N}$.
Perpeptual Bond Never matures, $P = \frac{cF}{\lambda}$.
Price Yield Relationship P is a decreasing function of λ, and the curve is convex (bowl shaped).
Macaulay Duration Weighted average time to maturity of the cash flow from a bond. $D = \frac{\sum_{i=1}^n t_i PV\{(c_i,t_i)\}}{PV(C)} = \sum_{i=1}^n w_i t_i$. If $c_i \geq 0$, then $t_1 \leq D \leq t_n$. For a zero coupon bond with maturity T, $D = T$.

Yield Curve Relationship between yield and maturity; Upward Sloping / Normal: During periods of economic expansion with low inflation, Downward Sloping / Inverted: Observed during or before recessionary periods or when inflation is high.
Spot Rate Annual interest rate for money held from today until some future time.
Term Structure of Interest Rates Collection of spot rates.
Rate Conventions Assume a yearly convention, so that $(1 + s_t)^t$ is the factor by which a deposit held for t years will grow.
Spot Rate Curve Plot of spot rates against time to maturity. Yield curve can be determined from spot rates.
Forward Rate Interest rate observed at some future time. $(1 + s_k)^k = (1 + s_j)^j(1 + f_{j,k})^{k-j}$ for $k > j > 0$.
Determining Spot Rate Yield to maturity of a zero coupon bond that matures at t. Then $P = \frac{R}{(1+s_n)^n}$, $s_n = (\frac{R}{P})^{1/n} - 1$.
Bootstrap Method Suppose we have determined previous spot rates. Then we estimate s_n using the n-period bond with cash flow $(-P_n, cF/m, ..., cF/m + R)$. Using the pricing formula $P_n = \frac{cF/m}{1+s_1} + ... + \frac{cF/m+R}{(1+s_n)^n}$, we can find s_n. Use the zero-coupon bond to estimate s_1, 2 period coupon paying bond for s_2, etc.



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